

LabProgram4 – Methods Report

1. Aim

To implement methods in Java through four problems: calculating the factorial of a number, checking whether a number is prime, finding the maximum of two numbers, and calculating the area of a circle.

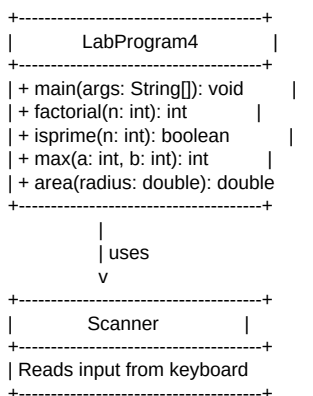
2. Description

The program is written in Java under the package `observationtasks` and uses the `Scanner` class to accept input. Four method-based practice problems are included. The first three programs are commented out, while the fourth program is active. Each problem defines a method that performs a specific calculation or check and returns a result to the `main()` method.

3. Tools Used

- Programming Language: Java
- Development Environment: Any Java IDE or text editor
- Execution Tool: Java Development Kit (JDK)
- Input Class: `java.util.Scanner`
- Concepts: static methods, parameters, return values, loops, conditional statements, and arithmetic operators

4. UML Diagram



Method concepts:

- static methods
- return values
- parameters
- method invocation

5. Algorithm

1. Start the program.
2. Import the `Scanner` class and create a `Scanner` object.
3. For Problem 1, read `n`, call `factorial(n)`, multiply the numbers from 1 to `n`, and return the factorial.
4. For Problem 2, read `n`, call `isprime(n)`, check divisibility from 2 to `n/2`, and return `true` or `false`.
5. For Problem 3, read two numbers, call `max(a, b)`, compare the values, and return the larger number.
6. For Problem 4, read the radius, call `area(radius)`, calculate $3.14 \times \text{radius} \times \text{radius}$, and return the area.
7. Display the returned result.

8. Stop the program.

6. Complete Java Source Code

```
package observationtasks;
import java.util.Scanner;

// Task on methods: 1st problem - factorial
/*
public class LabProgram4 {
    static int factorial(int n) {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }
        return fact;
    }

    public static void main(String[] args) {
        Scanner fa = new Scanner(System.in);
        int n, result;
        System.out.println("enter the number");
        n = fa.nextInt();
        result = factorial(n);
        System.out.println("factorial = " + result);
    }
}
*/

// MethodTask: 2nd problem - prime number
/*
public class LabProgram4 {
    static boolean isprime(int n) {
        if (n <= 1)
            return false;

        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }
        return true;
    }

    public static void main(String[] args) {
        Scanner pr = new Scanner(System.in);
        int n;
        System.out.println("enter the number");
        n = pr.nextInt();

        if (isprime(n))
            System.out.println("PRIME NUMBER");
        else
            System.out.println("not a prime number");
    }
}
*/

// MethodTask: 3rd problem - maximum of two numbers
/*
public class LabProgram4 {
    static int max(int a, int b) {
        if (a > b)
            return a;
        else
            return b;
    }

    public static void main(String[] args) {
        Scanner mt = new Scanner(System.in);
        int a, b, max;
        System.out.println("enter the values of a,b");
        a = mt.nextInt();
        b = mt.nextInt();
        max = max(a, b);
        System.out.println("maximum is " + max);
    }
}
*/

// MethodTask: 4th problem - area of a circle
public class LabProgram4 {
    static double area(double radius) {
        return 3.14 * radius * radius;
    }

    public static void main(String[] args) {
        Scanner sr = new Scanner(System.in);
```

```
double radius, result;

System.out.println("enter the radius");
radius = sr.nextDouble();
result = area(radius);
System.out.println("area = " + result);
}
}
```

7. Sample Output

Problem 1: Factorial

Input:

enter the number

5

Output:

factorial = 120

Problem 2: Prime Number

Input:

enter the number

7

Output:

PRIME NUMBER

Problem 3: Maximum of Two Numbers

Input:

enter the values of a,b

25 18

Output:

maximum is 25

Problem 4: Area of a Circle (Active Program)

Input:

enter the radius

5

Output:

area = 78.5

8. Result

The method-based problems were successfully demonstrated in Java. The programs illustrate how static methods can receive parameters, perform operations, and return values. The active fourth program calculates and displays the area of a circle using the entered radius.